

Introduction to Microsoft Azure: Describe Azure architecture and services

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Exercise 1: Explore interacting with Microsoft Azure

Objective

The goal of this exercise is to understand the different ways of interacting with Azure: the Azure Portal (graphical interface), Cloud Shell (PowerShell and Bash), and the Azure CLI interactive mode. You will learn how to switch between environments, run simple commands, and explore the `az` family of commands.

Step 1: Open the Azure Portal and Start Cloud Shell

- Open <https://portal.azure.com> in your browser and sign in.
- In the top-right action bar, click the **Cloud Shell** icon.
- Wait for Cloud Shell to initialize.

Step 2: Choose Mode — PowerShell or Bash

- PowerShell sessions show `PS` at the prompt.
- Bash sessions show `username@azure`.
- You can switch anytime with `pwsh` (for PowerShell) or `bash` (for Bash).

Step 3: PowerShell Quick Command

`Get-Date`

Step 4: Check Azure CLI Version

`az version`

Step 5: Bash-specific Command

`date`

Step 6: Update the CLI

`az upgrade`

Step 7: Interactive Mode

`az interactive`

- Respond to the telemetry prompt with YES or NO.
- Autocomplete suggestions appear after typing a few characters.
- Exit the session with `exit`.

Reflection

- Cloud Shell saves you from installing the CLI locally.
- Both PowerShell and Bash are supported and integrate with Azure.
- Interactive mode is especially helpful when discovering commands.

Exercise 2: Create an Azure Resource

In this exercise, you create a resource using the Azure portal. The main focus is to see how Azure resource groups populate with related resources.

Task 1: Create a Virtual Machine

1. Sign in to the Azure portal.
2. Select: **Create a resource** > **Compute** > **Virtual Machine** > **Create**.
3. Fill in the basics tab with the following details:

Setting	Value
Subscription	Your subscription
Resource group	New: IntroAzureRG
Virtual machine name	my-VM
Region	Default
Availability options	Default
Zone Options	Self-selected zone
Availability zone	Default
Security type	Default
Image	Default
Authentication type	Password
Username	azureuser
Password	Your password
Public inbound ports	None

4. Select **Review and Create**.
5. Review cost details (sandbox incurs no charges).
6. Select **Create** and wait for deployment.

Task 2: Verify Resources Created

1. Go to **Resource groups**.
2. Open **IntroAzureRG**.
3. Confirm the VM and related resources appear (network, storage, etc.).

Clean Up

1. Navigate to **Resource groups**.
2. Select **IntroAzureRG**.
3. Choose **Delete resource group**.
4. Type IntroAzureRG to confirm.

Exercise 3: Create an Azure Virtual Machine

Here, you create an Azure virtual machine (VM) and install Nginx as a web server. All steps are done through the Azure CLI in Cloud Shell.

Task 1: Create a Resource Group

```
az group create --name IntroAzureRG --location eastus
```

Task 2: Create a Linux Virtual Machine

```
az vm create \
  --resource-group "IntroAzureRG" \
  --name my-vm \
  --public-ip-sku Standard \
  --image Ubuntu2204 \
  --admin-username azureuser \
  --generate-ssh-keys
```

This provisions an Ubuntu 22.04 VM using SSH authentication.

Task 3: Install Nginx

```
az vm extension set \
  --resource-group "IntroAzureRG" \
  --vm-name my-vm \
  --name customScript \
  --publisher Microsoft.Azure.Extensions \
  --version 2.1 \
  --settings '{"fileUri":["https://raw.githubusercontent.com/MicrosoftDocs/mslearn-w"]}' \
  --protected-settings '{"commandToExecute": "./configure-nginx.sh}"'
```

This script updates the package list, installs Nginx, and configures the home page.

Exercise 4: Configure Network Access

In this exercise, you configure network access so your VM running Nginx is reachable from the internet. By default, only SSH is allowed. You will add an inbound rule for HTTP (port 80).

Task 1: Test Web Server Access

```
IPADDRESS="$(az vm list-ip-addresses \
  --resource-group "IntroAzureRG" \
  --name my-vm \
  --query "[] .virtualMachine.network.publicIpAddresses[*].ipAddress" \
  --output tsv)"
```

```
curl --connect-timeout 5 http://$IPADDRESS
```

The result times out, confirming port 80 is blocked.

Task 2: Check Security Rules

```
az network nsg list --resource-group "IntroAzureRG" --query '[] .name' --output tsv
```

Expected output: my-vmNSG

```
az network nsg rule list \
  --resource-group "IntroAzureRG" \
  --nsg-name my-vmNSG \
  --query '[] .{Name:name, Priority:priority, Port:destinationPortRange, Access:access}' \
  --output table
```

Default rule shows only SSH allowed.

Task 3: Add HTTP Rule

```
az network nsg rule create \  
  --resource-group "IntroAzureRG" \  
  --nsg-name my-vmNSG \  
  --name allow-http \  
  --protocol tcp \  
  --priority 100 \  
  --destination-port-range 80 \  
  --access Allow
```

Task 4: Retry Access

```
curl --connect-timeout 5 http://$IPADDRESS
```

Expected output:

```
<html><body><h2>Welcome to Azure! My name is my-vm.</h2></body></html>
```

Clean Up

Delete the resource group IntroAzureRG from the portal.

Exercise 5: Create a Storage Blob

This exercise walks through creating a storage account, adding a container, uploading a blob, and managing access.

Task 1: Create a Storage Account

- In the portal, select **Create a resource** → **Storage** → **Storage account**.
- Use subscription defaults, resource group IntroAzureRG, and a unique account name.
- Performance: Standard, Redundancy: LRS.
- On the Advanced tab, enable anonymous access for containers.
- Select **Review** then **Create**.

Task 2: Work with Blob Storage

- Under **Data storage**, open **Containers**.
- Add a container named **zama-container**, access level: Private.
- Upload a file (e.g., Lionel Messi wallpaper).

Task 3: Access Your Blob

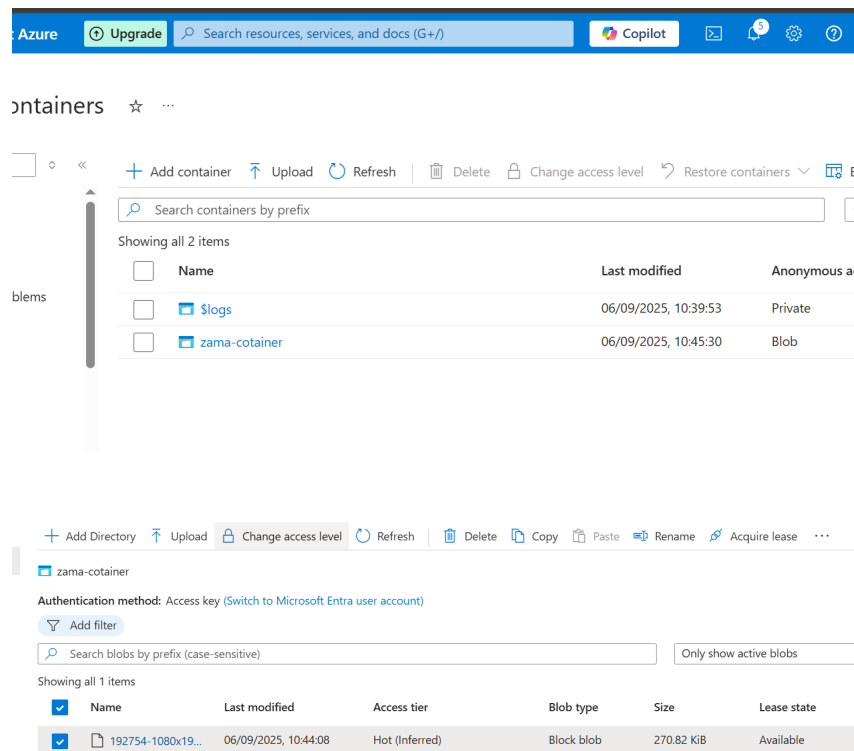
Blob URL: `https://olambda.blob.core.windows.net/zama-cotainer/192754-1080x1920-iphone-f.jpg`

At first, access returns:

```
<Error>
<Code>ResourceNotFound</Code>
<Message>The specified resource does not exist.</Message>
</Error>
```

Change access level to **Blob (anonymous read access for blobs only)** and refresh the page to see the image.

Screenshots



Clean Up

Remove all resources by deleting the IntroAzureRG group.